

Multiple Choice

1. **Answer:** A

Options: A) False, True; B) False, False and x is less than 100; C) False, True and x is an odd number; D) True, True and x is greater than 100; E) False, True and x is a prime number

Note: Correct option: A - False, True

2. **Answer:** A

Options: A) 0.0; B) Undefined; C) 2; D) $n-1$; E) $n+1$

Note: Correct option: A - 0.0

3. **Answer:** A

Options: A) $e/2$; B) 1; C) e^2 ; D) $2/e$; E) $1/2$

Note: Correct option: A - $e/2$

4. **Answer:** A

Options: A) 2.5; B) -2.0; C) 3.0; D) 2.0; E) 1.5

Note: Correct option: A - 2.5

5. **Answer:** C

Options: A) 42; B) 14; C) 35; D) 21; E) 49

Note: Correct option: C - 35

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '8'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** $(-\infty, -4) \cup (4, \infty)$. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 13.5 cm. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key